

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 8: Logarithms

Topic-wise short notes with solved examples

The meaning of a logarithm as an index, changing between exponential and logarithmic forms, the three laws of logarithms and their use in simplifying and solving.

1 Definition

- If $a^x = N$, where $a > 0$, $a \neq 1$ and $N > 0$, then x is the **logarithm of N to the base a** .

$$a^x = N \text{ means the same as } \log_a N = x$$

- A logarithm is an **index**: $\log_a N$ is the power to which a must be raised to get N .
- Logarithms are defined only for **positive** numbers. The base is positive and not 1.

Exponential form	Logarithmic form
$2^5 = 32$	$\log_2 32 = 5$
$10^3 = 1000$	$\log_{10} 1000 = 3$
$3^{-2} = \frac{1}{9}$	$\log_3 \frac{1}{9} = -2$
$9^{\frac{1}{2}} = 3$	$\log_9 3 = \frac{1}{2}$

- Logarithms to base 10 are **common logarithms**. When no base is written, base 10 is understood.

2 Basic results

- $\log_a 1 = 0$, because $a^0 = 1$.
- $\log_a a = 1$, because $a^1 = a$.
- $\log_a a^n = n$ and $a^{\log_a N} = N$.
- $\log_{10} 10 = 1$, $\log_{10} 100 = 2$, $\log_{10} 0.1 = -1$, $\log_{10} 0.01 = -2$.

Example 2.1 Find $\log_2 64$, $\log_5 \frac{1}{125}$ and $\log_{\sqrt{3}} 9$.

SOLUTION

$$2^6 = 64; 5^{-3} = \frac{1}{125}; (\sqrt{3})^4 = 9$$

6, -3 and 4

Example 2.2 Find x if $\log_x 81 = 4$, and if $\log_3 x = -2$.

SOLUTION

$$x^4 = 81 = 3^4, \text{ so } x = 3. \text{ And } x = 3^{-2}.$$

$$x = 3; x = \frac{1}{9}$$

Example 2.3 Find x if $\log_8 x = \frac{2}{3}$.

SOLUTION

$$x = 8^{\frac{2}{3}} = (2^3)^{\frac{2}{3}}$$

$$4$$

3 Laws of logarithms

- All logarithms in a law must have the same base.

Law	Statement
Product	$\log_a(mn) = \log_a m + \log_a n$
Quotient	$\log_a \frac{m}{n} = \log_a m - \log_a n$
Power	$\log_a m^n = n \log_a m$

- They follow from the laws of indices: if $m = a^x$ and $n = a^y$ then $mn = a^{x+y}$.
- There is **no** law for $\log(m+n)$ or $\log(m-n)$.

Example 3.1 Express $\log \frac{a^2 b}{\sqrt{c}}$ in terms of $\log a$, $\log b$ and $\log c$.

SOLUTION

$$\log a^2 + \log b - \log c^{\frac{1}{2}}$$

$$2 \log a + \log b - \frac{1}{2} \log c$$

Example 3.2 Write as a single logarithm: $2 \log 3 + \log 5 - \log 9$.

SOLUTION

$$\log 9 + \log 5 - \log 9$$

$$\log 5$$

Example 3.3 Evaluate $\log 25 + \log 4$ and $\log_2 96 - \log_2 3$.

SOLUTION

$$\log 100 = 2 \text{ and } \log_2 32 = 5$$

2 and 5

Example 3.4 Simplify $\frac{\log 27}{\log 9}$.

SOLUTION

$$\frac{3 \log 3}{2 \log 3}$$

$$\frac{3}{2}$$

4 Using given values

- Break each number into factors whose logarithms are known.
- $\log 5$ can be found from $\log 2$: $\log 5 = \log \frac{10}{2} = 1 - \log 2$.

Example 4.1 Given $\log 2 = 0.3010$ and $\log 3 = 0.4771$, find $\log 12$, $\log 5$ and $\log 1.5$.

SOLUTION

$$\log 12 = 2 \log 2 + \log 3 = 0.6020 + 0.4771$$

$$\log 5 = 1 - 0.3010$$

$$\log 1.5 = \log 3 - \log 2$$

1.0791, 0.6990 and 0.1761

Example 4.2 If $\log 2 = x$ and $\log 3 = y$, express $\log 72$ and $\log 4.5$ in terms of x and y .

SOLUTION

$$72 = 2^3 \times 3^2 \text{ and } 4.5 = \frac{9}{2}$$

$3x + 2y$ and $2y - x$

5 Equations with logarithms

- Combine the logarithms on each side into one, using the laws.
- If $\log_a m = \log_a n$, then $m = n$.
- Change a number into a logarithm when needed: $1 = \log 10$, $2 = \log 100$.
- **Reject** any value that makes the number inside a logarithm zero or negative.

Example 5.1 Solve $\log(x + 3) + \log(x - 3) = \log 16$.

SOLUTION

$\log(x^2 - 9) = \log 16$, so $x^2 = 25$ and $x = \pm 5$.

$x = -5$ makes $x + 3$ negative.

$$x = 5$$

Example 5.2 Solve $\log x + \log 5 = 2$.

SOLUTION

$$\log 5x = \log 100$$

$$x = 20$$

Example 5.3 Solve $2 \log x - \log(x + 6) = 0$.

SOLUTION

$$\log \frac{x^2}{x + 6} = \log 1, \text{ so } x^2 - x - 6 = 0 \text{ and } x = 3 \text{ or } x = -2.$$

$$x = 3$$

Example 5.4 If $\log(a + b) = \log a + \log b$, express a in terms of b .

SOLUTION

$$a + b = ab, \text{ so } a(b - 1) = b$$

$$a = \frac{b}{b - 1}$$

Example 5.5 If $a^2 + b^2 = 23ab$, show that $\log \frac{a + b}{5} = \frac{1}{2}(\log a + \log b)$.

SOLUTION

$$(a + b)^2 = 25ab, \text{ so } \left(\frac{a + b}{5}\right)^2 = ab.$$

$$\text{Taking logarithms: } 2 \log \frac{a + b}{5} = \log a + \log b$$

6 Common mistakes

- Writing $\log(m + n) = \log m + \log n$ or $\log(m - n) = \log m - \log n$.
- Writing $\frac{\log m}{\log n} = \log m - \log n$. The quotient law is for $\log \frac{m}{n}$.
- Writing $(\log m)^n = n \log m$. The power law is for $\log m^n$.
- Keeping a negative solution that makes a logarithm undefined.
- Forgetting that $\log 1 = 0$, not 1.
- Mixing bases inside one law.